Lecturers: Thomas Quella and Nick Beaton
School of Mathematics and Statistics

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MAST20030: Differential Equations
Tutorial Solutions: Week 11

Solution to Question 1.

(a) The initial-boundary value problem takes the following form:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \qquad \begin{matrix} u(0,t) = 0, \\ u(1,t) = 0, \end{matrix} \qquad u(x,0) = \begin{cases} \dfrac{A}{\ell} x & \text{if } 0 \leqslant x \leqslant \ell, \\ \dfrac{A}{1-\ell}(1-x) & \text{if } \ell \leqslant x \leqslant 1, \end{cases} \qquad \partial_t u(x,0) = 0.$$

(b) Separation of variables, $u(x,t) = X(x)T(t)$, converts the wave equation into

$$\ddot{T}(t) + \lambda c^2 T(t) = 0 \qquad \text{and} \qquad X''(x) + \lambda X(x) = 0,$$

as usual. The boundary conditions imply Dirichlet conditions in x, *ie.* $X(0) = X(1) = 0$, hence the eigenvalues and eigenfunctions are

$$\begin{aligned} &\lambda_n = \pi^2 n^2, \qquad X_n(x) = \sin(\pi n x) \qquad (n = 1, 2, 3, \ldots) \\ \Rightarrow \qquad &T_n(x) = A_n \cos(\pi n c t) + B_n \sin(\pi n c t) \qquad (A_n, B_n \in \mathbb{C}). \end{aligned}$$

Imposing the homogeneous initial condition then requires $\dot{T}_n(0) = 0$, which forces $B_n = 0$. (In principle, the initial condition needs to be imposed on the general solution, not on any individual $T_n(t)$. Convince yourself that this procedure leads to the same conclusion in the present case!)

The separated solutions of the (homogeneous and linear) wave equation with the homogeneous (and linear) boundary and initial conditions are therefore $X_n(x)T_n(t) = A_n \sin(\pi n x)\cos(\pi n c t)$. Homogeneity and linearity then implies that linear combinations of these separated solutions are also solutions satisfying the homogeneous boundary and initial conditions. We therefore guess that the solution has the form

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin(\pi n x)\cos(\pi n c t),$$

where we have relabelled A_n as b_n.

Imposing the inhomogeneous initial condition gives

$$u(x,0) = \sum_{n=1}^{\infty} b_n \sin(\pi n x)$$

$$\begin{aligned} \Rightarrow \; b_n &= 2\int_0^1 u(x,0)\sin(\pi n x)\,\mathrm{d}x \\ &= 2\int_0^\ell \frac{A}{\ell} x \sin(\pi n x)\,\mathrm{d}x + 2\int_\ell^1 \frac{A}{1-\ell}(1-x)\sin(\pi n x)\,\mathrm{d}x \\ &= \frac{2A}{\pi^2 n^2}\frac{\sin(\pi n \ell)}{\ell(1-\ell)} \qquad \text{(integrate by parts).} \end{aligned}$$

We record, for what follows, that

$$\left|\frac{b_n}{b_1}\right| = \frac{1}{n^2}\frac{|\sin(\pi n \ell)|}{\sin(\pi \ell)}.$$

(c) If we set $\ell = \frac{1}{2}$, so we are plucking in the middle of the string, then we obtain

$$b_n = \frac{8A}{\pi^2 n^2}\sin\frac{\pi n}{2} = \begin{cases} \dfrac{8A(-1)^m}{\pi^2(2m+1)^2} & \text{if } n = 2m+1 \text{ is odd}, \\ 0 & \text{if } n \text{ is even}. \end{cases}$$

The first six ratios $|b_n/b_1|$ are thus

n	1	2	3	4	5	6
$\lvert b_n/b_1\rvert$	1	0	$\frac{1}{9}$	0	$\frac{1}{25}$	0

(d) The tabulations $\ell = \frac{1}{3}$ are

$$\ell = \tfrac{1}{3}$$

n	1	2	3	4	5	6
$\lvert b_n/b_1\rvert$	1	$\frac{1}{4}$	0	$\frac{1}{16}$	$\frac{1}{25}$	0

Taking $\ell = \frac{1}{n}$ basically kills the n-th harmonic — the ratio b_n/b_1 is zero. It also kills all multiples of the n-th harmonic — the ratios b_{jn}/b_1 are zero for all $j = 1, 2, 3, \ldots$

(e) l'Hôpital's rule gives

$$\lim_{\ell\to 0^+} b_n = \lim_{\ell\to 0^+} \frac{2A}{\pi^2 n^2}\frac{\sin(\pi n\ell)}{\ell(1-\ell)} = \frac{2A}{\pi n}, \qquad \lim_{\ell\to 0^+}\left|\frac{b_n}{b_1}\right| = \left|\lim_{\ell\to 0^+}\frac{1}{n^2}\frac{\sin(\pi n\ell)}{\sin(\pi\ell)}\right| = \frac{1}{n}.$$

In this limit, all harmonics contribute and the ratios decay as $\frac{1}{n}$ instead of $\frac{1}{n^2}$. The greater importance of the higher harmonics is responsible for the harsher sound that results from plucking near the ends.